

Program overview

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Year 2024/2025
Organization Applied Sciences
Education Bachelor Applied Physics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Bachelor TN 2024							
1e jaar BSc TN 2024							
TN1003	Design Engineering voor Fysici	6					
TN1101	Technology Management	3					
TN1142WI	Linear Algebra part 1	3					
TN1201	Thermodynamics	3					
TN1401WI	Analyse voor TNW 1	6					
TN1402WI	Analyse voor TNW 2	6					
TN1405-P	Inleidend Practicum	6					
TN1410WI	Advanced Calculus	3					
TN1612TU	Mechanics and Theory of Relativity	6					
TN2054	Electromagnetism	6					
1e jaar BSc TN 2024 - module							
TN1302	Introduction to Mechanics, Waves and Optics	3					
TN1312	Modern Physics	3					
TN1322	Introduction to Electricity and Magnetism	3					
1e jaar BSc TN 2024 - keuzevak							
TN1651	Introduction to Biophysics	3					
TN1851	Nuclear Science and Technology	3					
TN2811	Introduction to Elementary Particle Physics	3					
2e jaar BSc TN 2024							
TN2003	Design Engineering voor Fysici 2	6					
TN2211	Electronic Instrumentation	6					
TN2244WI	Linear Algebra an Differential Equations	6					
TN2321	Classical Mechanics	3					
TN2345	Introduction to Waves	3					
TN2513	Computational Science	3					
TN2545	Systems and Signals	6					
TN2626	Statistical Physics	6					
TN2786	Physical Transport Phenomena	6					
TN2953-P	Research Laboratory	6					
TN3104WI	Probability and Statistics	3					
2e jaar Module 2024							
TN2304	Quantum Mechanics 1	3					
TN2314	Quantum Mechanics 2	3					
3e jaar BSc TN 2024							
SC3011TN	Stochastic Signal Analysis	3					
TN2412	Quantum Mechanics 3	3					
TN2421	Optics	3					
TN2844	Solid State Physics	6					
TN2983	Bachelor Thesis	12					
TN3100	Individual research project (var EC)	0					
WM0318TN	Argumentation and Philosophy of Science	3					

Year

Organization

Education

2024/2025

Applied Sciences

Bachelor Applied Physics

Bachelor TN 2024

Year	2024/2025
Organization	Applied Sciences
Education	Bachelor Applied Physics

1e jaar BSc TN 2024

TN1003	Design Engineering voor Fysici	6
Responsible Instructor	Dr.ir. R.W. Hut	
Instructor	R. Ockhorst	
Education Period	3A 3B 4A 4B	
Start Education	3A	
Exam Period	3A 4A	
Course Language	Dutch	

TN1101	Technology Management	3
Responsible Instructor	Dr. M. Sovago	
Instructor	Dr. K.W.A. van Dongen	
Education Period	3A 3B 4A 4B	
Start Education	3A	
Exam Period	Different, to be announced	
Course Language	English	

TN1142WI	Linear Algebra part 1	3
Responsible Instructor	Dr. S.K.J. Aerts	
Instructor	Dr. P.M. Visser	
Education Period	4A	
Start Education	4A	
Exam Period	4A 5AB	
Course Language	Dutch	
Assessment	Het tentamen is on campus, schriftelijk met mogelijk een digitale component op eigen of beheerde laptop en heeft 1 herkansing.	

TN1201	Thermodynamics	3
Responsible Instructor	Prof.dr.ir. C.R. Kleijn	
Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	4x 2 uur p/w hoorcollege 4x 2 uur p/w werkcollege	
Education Period	4B	
Start Education	4B	
Exam Period	4B 5AB	
Course Language	Dutch	
Course Contents	1. Basic concepts: open and closed systems; intensive and extensive properties; 2. Energy; First law of Thermodynamics; heat and work; energy balance for closed systems; 3. Properties; p-v-T relations for generic media, incompressible media and ideal gases; internal energy and enthalpy; 4. Energy balance for open systems; 5. Second law of thermodynamics; thermodynamics cycles; reversible and irreversible processes; Carnot cycle; 6. Entropy; Entropy balance for closed and open systems; isentropic processes; isentropic efficiency	
Study Goals	1. Acquire basic knowledge of phenomenological thermodynamics, including its first and second law. 2. Acquire insight in the relationship between the laws of phenomenological thermodynamics and the possibilities and limitations of technological solutions for societal problems. 3. Be able to independently acquire knowledge in the field of phenomenological thermodynamics. 4. Be able to solve standard problems within the field of phenomenological thermodynamics, in relation to broader technological applications, making use of analytical theoretical methods, at the level of the problems in the internationally widely used book by Morand and Shapiro. 5. Be able to solve novel problems within the field of phenomenological thermodynamics, in relation to broader technological applications, making use of analytical theoretical methods, at the level of the problems in the internationally widely used book by Morand and Shapiro.	
Education Method	semi-block teaching; 5 weeks, 16 hours/week. Lectures, self-study, demonstrations, tests, exercises, intermediate exams.	
Literature and Study Materials	M.J. Moran, H.J. Shapiro, D.D. Boettner & M.B. Bailey, Moran's Principles of Engineering Thermodynamics, SI Version, Global Edition, Wiley, 2017, ISBN 978-1-119-45406-9	
Books	M.J. Moran, H.J. Shapiro, D.D. Boettner & M.B. Bailey, Moran's Principles of Engineering Thermodynamics, SI Version, Global Edition, Wiley, 2017, ISBN 978-1-119-45406-9	
Assessment	Non-mandatory midterm exam on chapter 1+2+3+4: 30%; Non-mandatory 13 minitests: together 20%; Mandatory final exam on chapters 1-6 (50-100%, depending on scores for midterm and minitests). Re-sit exam on chapters 1-6 (100%).	
Permitted Materials during Tests	tables (will be provided); simple, non-graphical calculator; formula sheet (will be provided)	
Studyload/Week	16 hours/week during 5 weeks.	

TN1401WI	Analyse voor TNW 1	6
Responsible Instructor	Dr. P.M. Visser	
Instructor	Drs. A.T. Hensbergen	
Instructor	Dr. S.K.J. Aerts	
Education Period	1A 1B	
Start Education	1A	
Exam Period	1A 1B 2A	
Course Language	Dutch	
Tags	Analysis Calculus	

TN1402WI	Analyse voor TNW 2	6
Responsible Instructor	Dr. P.M. Visser	
Instructor	Dr. B. van den Dries	
Instructor	Dr. S.K.J. Aerts	
Education Period	2A 2B	
Start Education	2A	
Exam Period	2A 2B 3A	
Course Language	Dutch	
Tags	Analysis Calculus	

TN1405-P	Inleidend Practicum	6
Responsible Instructor	Dr. C.F.J. Pols	
Course Coordinator	Dr. C.F.J. Pols	
Instructor	Dr. A. Mohammadi Gheidari	
Education Period	1A 1B 2A 2B	
Start Education	1A	
Exam Period	Different, to be announced 1B 2A	
Course Language	Dutch	
Course Contents	Python Data-analysis, Measurement and uncertainty Introductory experiments 1-2 Development experiment 1-3 Final project	
Study Goals	In this course we lay the foundation for experimental physics. We especially focus on collection, processing and analysing data, calculating how errors progress, representing data using python and presenting the findings by means of a written report.	
Education Method	Instruction Experimental work Programming	
Assessment	Python assignments (week 3&5) (10%) Toets python en data-analyse (herkansbaar) (20%) Verslag IE1 (15%) Boltzmann (10%) RC experiment (10%) Bepaling van g (10%) Eindproject (25%) Het eindcijfer wordt pas in OSIRIS verwerkt (en studiepunten dus toegekend) wanneer voor alle onderdelen een cijfer genoteerd staat, en elk cijfer minstens een 5.5 is. Er is dus voor elk onderdeel een inspanningsverplichting. Wanneer een onderdeel onvoldoende is afgerond, moet aan het eind van de 5e week waarin het cijfer is gepubliceerd het werk verbeterd en ingeleverd zijn. Wanneer alle punten van feedback voldoende zijn verbeterd, wordt het cijfer 5.5 toegekend. Het advies is dan ook het werk meteen goed te doen en je bij een TA te melden wanneer je merkt dat het je niet lukt om de gevraagde kwaliteit te leveren.	

TN1410WI	Advanced Calculus	3
Responsible Instructor	Dr. B. van den Dries	
Instructor	Dr. S.K.J. Aerts	
Education Period	3A 3B	
Start Education	3A	
Exam Period	3A 3B 5AB	
Course Language	Dutch	

TN1612TU	Mechanics and Theory of Relativity	6
Responsible Instructor	Prof.dr. B. Rieger	
Instructor	Prof.dr. R.F. Mudde	
Contact Hours / Week x/x/x/x	4x 2 uur hoorcollege 4x 2 uur werkcollege	
Education Period	3A 3B	
Start Education	3A	
Exam Period	3A 3B 4B	
Course Language	Dutch	
Course Contents	Klassieke Mechanica Wetten van Newton en Galilei-transformaties; behoud van energie, impuls en impulsmoment; centrale krachtvelden, wetten van Kepler; stelsels van puntmassas, dynamica van starre lichamen, beweging in niet-inertiaal systemen, schijnbare krachten; botsingen;	
Study Goals	Speciale Relativiteit Einstein's Axioma's, Lorentz transformaties, tijdrek & lengtekrimp, ruimte-tijd geometrie, relativistische kinematica & Doppler effect, vier-vectoren, massa en energie equivalentie, relativistische dynamica. - Basiskennis verkrijgen van de basiswetten van de klassieke mechanica [1] - Basiskennis verkrijgen van de speciale relativiteitstheorie en van de beperkingen van de wetten van Newton [1] - Basisproblemen van de klassieke mechanica kunnen analyseren [2,3,4] - Basisproblemen van de special relativiteitstheorie kunnen analyseren [2,3,4] - Basisproblemen van de klassieke mechanica kunnen oplossen [2,3,4] - Basisproblemen van de special relativiteitstheorie kunnen oplossen [2,3,4]	
Education Method	3e kwartaal: vier dagdelen, steeds startend met 1e uur college (theorie & demos), gevolgd door zelfstudie en begeleide oefenopgaven.	
Literature and Study Materials	W.D. McComb, Dynamics and Relativity, Oxford University Press. Gehele boek, behalve hoofdstukken 8 en 15. Alternatief: J.R. Taylor, Classical Mechanics, University Science Books. Geslecteerde hoofdstukken zoals aangegeven tijdens college. https://qiweb.tudelft.nl/mecharela/	
Assessment	2 Schriftelijk bonustoetsen (BT). Deze tellen voor maximaal 20% in totaal. Een betere score op het eindtentamen prevaleert altijd boven de bonustoetsen. Schriftelijk eindtentamen (ET) hele stof in 10e week. Als ET > alle BT of geen BT gemaakt final grade = ET Als 2 BT > ET final grade = 24/30 ET + 6/30 (BT1 + BT2) Als 1 BT > ET final grade = 27/30 ET + 3/30 BT	

TN2054	Electromagnetism	6
Responsible Instructor	Dr.ir. M.D. Verweij	
Instructor	Dr. J. Kalkman	
Instructor	G.G.J. Renaud	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	4A 4B	
Start Education	4A	
Exam Period	4B 5AB	
Course Language	Dutch	
Course Contents	Electrostatics in vacuum and matter (including: electric field, electric potential, Gauss's law, mathematical techniques), magnetostatics in vacuum and matter (including: magnetic field, vector potential, Ampère's law), electrodynamics (including: induction and Maxwell's equations), electromagnetic waves in vacuum and matter.	
Study Goals	1. Being able to work with vector fields; acquiring knowledge about the fundamentals of electrostatics; being able to independently solve electrostatic problems. 2. Acquiring fundamental insight in the concepts of electrical fields and potentials in matter; being able to apply this theory to specific problems. 3. Acquiring knowledge about the fundamentals of magnetostatics; being able to independently solve magnetostatic problems. 4. Acquiring knowledge of the fundamentals of electrodynamics; understanding the basics and derivation of Maxwell's equations and electromagnetic waves, and being to apply this to specific problems.	
Education Method	Three times per week 2 hour lecture followed by 2 hours of self-study/exercises/instruction. In addition, three intermediate tests are offered.	
Literature and Study Materials	D.J. Griffiths, Introduction to Electrodynamics (5th edition), Cambridge University Press: Chapters 1-7 and parts of chapters 8 and 9 (see Brightspace).	
Assessment	Written optional test every two weeks (3 in total), and final written exam	

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1e jaar BSc TN 2024 - module

TN1302	Introduction to Mechanics, Waves and Optics	3
Responsible Instructor	Dr. H.J. Geertsema	
Instructor	Prof.dr. L. Kuipers	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	Dutch	

TN1312	Modern Physics	3
Responsible Instructor	Prof.dr. Y.M. Blanter	
Instructor	Dr. R.H.M. Smit	
Education Period	2A	
Start Education	2A	
Exam Period	2A 3A	
Course Language	Dutch	
Study Goals	- To qualitatively understand phenomena of modern physics - To produce simple quantitative estimates and to assess whether their order of magnitude is correct - To get an impression what a contemporary physics research is about - To prepare for coming years in which this way of working is more common	
Assessment	1) Non-mandatory homework assignments (weight 15% if higher than the exam grade); 2) Three non-mandatory tests (weight 5% for each test if higher than the exam grade); 3) Final exam for the whole material of the course (70%). The bonus points are not valid for the retake, which counts for 100%.	
Permitted Materials during Tests	Closed book exam in octal 3 and closed book retake in the spring. The list of formulas will be distributed during the exam. Calculators may not be used during the exam or tests.	

TN1322	Introduction to Electricity and Magnetism	3
Responsible Instructor	Dr. W.G. Bouwman	
Instructor	Prof.dr. R.F. Mudde	
Instructor	Prof.dr.ir. J.P. Hoogenboom	
Education Period	1B	
Start Education	1B	
Exam Period	1B 2B	
Course Language	Dutch	
Course Contents	Elektrische velden, de wet van Coulomb, potentiaal, capaciteit, dielectrica. Elektrische stromen, elektrische weerstand, netwerken. Magnetische velden, de wet van Biot en Savart, de wet van Ampere, magnetische flux, magnetische materialen, krachten op ladingen en kringstromen. Elektromagnetische inductie, de wet van Faraday, L-R en L-R-C netwerken, wisselstromen. Wetten van Maxwell en electromagnetische golven.	
Study Goals	1. Studenten passen de wetten van Maxwell en andere basiswetten van elektromagnetisme toe om eenvoudige problemen op te lossen: a) Om een grootheid uit te rekenen b) Om een (meer toegepaste) uitdrukking voor een grootheid te berekenen 2. Studenten kunnen de validiteit van een redenering evalueren gebaseerd op de basiswetten van het electromagnetisme. 3. Studenten begrijpen afleidingen om uit Maxwells wetten tot elektromagnetische golven te komen. 4. Studenten passen symmetrie toe om een wiskundige weergave in integraal of vectornotatie van een probleem op te lossen.	
Education Method	Hoorcolleges afgewisseld met korte video-instructies en (huiswerk)opdrachten via Mastering Physics.	
Assessment	Het eindtentamen is een schriftelijke toets. Daarnaast zijn er tussentoetsen die deels uit schriftelijke en deels uit digitale vragen bestaan. De tussentoetsen kunnen als bonus gelden.	

Year	2024/2025
Organization	Applied Sciences
Education	Bachelor Applied Physics

1e jaar BSc TN 2024 - keuzevak

TN1651	Introduction to Biophysics	3
Responsible Instructor	Dr. M.S. Bauer	
Instructor	Dr. J.W. Zwanikken	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 1x 2 uur p/w werkcollege	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Course Contents	The lectures provide an introduction to biophysics, on the levels of molecules, cells and complex organisms and communities. We will obtain an overview of important processes that sustain life on the cellular scale, as well as the experimental tools to investigate them and the mathematical/physics concepts that describe them. Biophysics is a growing and interdisciplinary physics subfield, with exciting theoretical as well as experimental research directions that help us appreciate the complexity of life of our planet. Current research topics in biophysics range from determining the intricate motion and function of proteins inside cells, cellular substructure and patterns, and cellular communication from neurons over cellular communities and larger animals. In order to form a quantitative understanding of these topics, we will explore the physical description of these systems: we will learn about probability distributions, ODEs and simulation techniques to calculate expectations, for small and large number of particles, reaction-diffusion equations and non-linear dynamics approaches to solve them, and end with an understanding of entropy. We will relate these descriptions to traditional topics in the physics curriculum, such as (stochastic) thermodynamics, non-linear dynamics and statistical physics. Topics include: single-molecule biophysics, molecular diffusion & random walk, simulation tools, probabilistic and stochastic descriptions of molecules, fluorescence microscopy & spectroscopy, neuronal information transmission including ion channels & electrophysiology, signaling pathways, reaction equations and their importance in development, stochasticity in gene expression, pattern formation, and biological computations.	
Study Goals	- Understand the key concepts in molecular and cellular biophysics. - Can interpret data from biophysical experiments and link the results to the studied concepts. - Can use physical insights to solve new problems related to biophysics.	
Education Method	The course is taught in English. It consists of lectures and in-class exercises.	
Literature and Study Materials	The course will not follow a single textbook. All study materials and lecture notes will be provided to the students during the course. A relevant textbook is: Biological Physics : Energy, Information, Life, door Philip Nelson W H Freeman & Co.	
Assessment	30% Participation grade (including required participation and presentation in a brief journal club providing an overview over recent biophysics work, as well as two out of four in-class (bonus grade) quizzes); 70% exam grade	

TN1851	Nuclear Science and Technology	3
Responsible Instructor	Dr.drs.ir. L.J. Bannenberg	
Instructor	Dr.ir. M. Rohde	
Instructor	Dr.ir. M.C. Goorden	
Education Period	2B	
Start Education	2A	
Exam Period	2B 3B	
Course Language	Dutch	
Study Goals	By the end of this course, you should be able to: - Reproduce theoretical knowledge about the physics of radiation. - Apply the theoretical knowledge about the physics of radiation to solve (simplified) problems related to radiation. - Explain how radiation can be used for medical applications including imaging and treatment. - Explain how nuclear reactors work. - Explain how and when neutron and X-ray radiation can be used to study materials using diffraction and imaging.	
Education Method	Plenary lectures, exercise lectures and excursions to the Reactor Institute Delft. The tour may take place after the educational period for practical reasons.	
Books	All course material is covered in the lecture slides	
Assessment	Examination consist of two part: three bonus quizzes and a written exam. The mean grade of the bonus quizzes counts for 15% towards the final course grade if and only if the mean grade of the bonus quizzes exceeds the grade for the final written exam. In all other cases the grade of the written exam counts for 100% towards the final grade of the course. The grade of the bonus quizzes is not valid for the resit or during the next academic year.	

TN2811	Introduction to Elementary Particle Physics	3
Responsible Instructor	Prof.dr. S.C.M. Bentvelsen	
Instructor	L. Brenner	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 2x 2 uur p/w werkcollege	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch English	
Course Contents	The course includes an introduction to the Standard Model of elementary particles, an introduction of various accelerator systems and the detection of elementary particles with different methods.	
Study Goals	Being able to estimate energies and distance scales that affect physics Introduction to modern subatomic physics Knowledge of quarks as building blocks of particles in the atomic nucleus Understanding the relationships between quarks, (charged) leptons and neutrinos Knowledge about the fundamental forces - insight into particles as a force carrier Knowledge of hadrons like mesons and baryons Understanding Special Relativity Knowledge of particle accelerators Knowledge of the interaction between fast charged particles and matter. Knowledge of the interaction of photons with matter Knowledge of modern detectors and detection techniques Insight into the open questions and current problems in subatomic and astroparticle physics Insight into physics beyond the standard model	
Education Method	Lectures and assignments	
Books	You can order the reader from EDUweb for TN2811 Elementaire Deeltjes Fysica yourself at http://www.uniportaal.nl/site At this website, please log in with your TU Delft NEtID and password. The reader TN2811 Elementaire Deeltjes Fysica can be found under 'faculteit TNW (toegepaste natuurwetenschappen), opleiding Technische Natuurkunde'. Additional information: Bestelnummer 06917560003. PRN/readernummer PRN00049717. Costs are 34,25.	
Assessment	Written exam	
Remarks	The course will be given by colleagues from Nikhef, Prof. dr. Stan Bentvelsen, Dr. Panos Christakoglou	

Year	2024/2025
Organization	Applied Sciences
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2e jaar BSc TN 2024

TN2003	Design Engineering voor Fysici 2	6
Responsible Instructor	Dr. W.G. Bouwman	
Instructor	Dr.ir. R.W. Hut	
Education Period	1A 1B	
Start Education	1A	
Exam Period	none	
Course Language	Dutch	

TN2211	Electronic Instrumentation	6
Responsible Instructor	Dr.ir. D.R. Schaart	
Instructor	I.Z. Esmaeil Zadeh	
Education Period	1A 1B 2A 2B 3A 3B 4A 4B	
Start Education	1A 3A	
Exam Period	2B 4B 5AB	
Course Language	Dutch	

TN2244WI	Linear Algebra an Differential Equations	6
Responsible Instructor	Drs. A.T. Hensbergen	
Responsible Instructor	Dr. S.K.J. Aerts	
Education Period	1A 1B	
Start Education	1A	
Exam Period	1A 1B 2B	
Course Language	Dutch	
Course Relations	Quantum Mechanics; Systems and Signals	
Tags	Abstract Adventurous Algebra Analysis	

TN2321	Classical Mechanics	3
Responsible Instructor	Prof.dr. S. Stallinga	
Instructor	Prof.dr. J.M. Thijssen	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2A	
Start Education	2A	
Exam Period	2A 3A	
Course Language	Dutch	
Expected prior knowledge	1. Mechanics from the first year of the physics bachelor programme 2. Mathematics from the first and second year of the physics bachelor programme, in particular ordinary differential equations and eigenvalue problems.	
Course Contents	Variational principle, degrees of freedom and constraints, Euler-Lagrange formalism, Hamilton formalism, Noether's theorem, coupled oscillators and normal modes	
Study Goals	The student who completes this course successfully 1. can formulate functionals for appropriate optimization problems and use the Euler equations to obtain their solution 2. can choose generalised coordinates for the description of classical mechanical problems possibly subject to constraints, formulate the Lagrangian or Hamiltonian for these and arrive at meaningful solutions using the Euler Lagrange or Hamilton equations 3. can identify conserved quantities in mechanical systems and relate this conservation property to continuous symmetries. 4. can approximate a system close to equilibrium and from there find the normal mode frequencies and amplitude vectors of the vibrations	
Education Method	Lectures and exercise classes. There is an emphasis on solving problems independently.	
Literature and Study Materials	'Classical Mechanics' by John R. Taylor	
Assessment	Assessment consists of a final written exam.	

TN2345	Introduction to Waves	3
Responsible Instructor	Dr. K.W.A. van Dongen	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch	
Course Contents	No English version, see Dutch version	
Study Goals	No English version, see Dutch version	
Education Method	No English version, see Dutch version	
Assessment	No English version, see Dutch version	

TN2513	Computational Science	3
Responsible Instructor	E. Greplová	
Instructor	Prof.dr. G.A. Steele	
Contact Hours / Week x/x/x/x	2x 4 uur p/w practicum	
Education Period	3A 3B	
Start Education	3A	
Exam Period	3B 4B	
Course Language	English	
Course Contents	Basics Python programming (review), numerical differentiation and integration, numerical solutions of equations, linear algebra, random numbers, Fourier analysis, solving ordinary and partial differential equations.	
Study Goals	Overall Learning Objectives (tested in Assignments and Exam): ===== * You are able to code solutions to problems that arise in physics using numerical techniques in Python * You are able to interpret the results your code produces in the context of the physics problem * You are able to identify how the implementation of your code affects the ability of the code to accurately represent the physical problem you are trying to solve Detailed learning objectives (taught and formatively tested in Lecture Notebooks): ===== Student is able to: * Write a function to calculate the numerical derivative of an arbitrary function using both the forward difference and the central difference approximation * Interpret the error made in central and forward numerical derivatives * Calculate the numerical derivative of a dataset * Implement integration of a function with the trapezoidal rule * Implement integration of a function with Simpson's rule * Calculate the integral of a numerical dataset * Find the root of a one-variable function with the bisection method * Find the root of a one-variable function with Newton's method * Solve a set of linear equations numerically * Explain when a set of linear equations can be solved * Determine eigenvalues and eigenvectors of a matrix * Calculate the Fast Fourier transform (FFT) of a dataset * Calculate the correct frequency vector for the transformed data * Convert the Fourier-transformed data into a physically meaningful power spectrum with the correct units * Use the FFT to determine the frequency and amplitude of weak periodic signals in noisy data * Use the FFT to filter numerical data * Interpret artifacts in your filtered data arising from the FFT * Calculate the 2D Fourier transform of 2D data * Construct 2D images of mask patterns and calculate the far-field diffraction pattern * Make random numbers and know how to analyze them * Implement your own code for numerical integration of a first order ordinary differential equation (ODE) using Euler's method * Implement numerical integration of coupled and second-order differential equations * Identify failure of the convergence of Euler integration * Implement your own code for second-order and fourth-order Runge-Kutta integration of ODEs * Implement solutions of ODEs using the solve_ivp() routine from the scipy library * Apply numerical integration to a physical problem and interpret your results * Use numerical integration to find the steady-state of the driven damped Harmonic oscillator * Solve boundary-value problems using the "shooting method" * Use numerical integration to solve problems with nonlinear damping * Implement the Jacobi and Gauss-Seidel methods of solving partial differential equations (PDEs) * Speed up iterative solution of PDDs by implementing successive over-relaxation (SOR)	

Education Method	* Solve for the electrostatic potential for 2-dimensional (2D) problems using SOR including fixed charges * Implement the FTCS method for solving initial-value PDEs * Solve and interpret the results from simulations of the 1-dimensional diffusion equation using FCTS * Identify the failure of the FTCS method for solving some problems due to numerical instability * Implement the spectral method for solving initial-value PDEs * Solve and interpret the results from simulations of the wave equation using the spectral method
	Online, self-paced course with Jupyter Notebooks that contain learning material, example code, and autograded exercises delivered in a cloud-based platform. Formative assignments that cover the implementation of the numerical techniques in the context of a physical problem will be supplied. These assignments demonstrate the expectations of what will be asked of the students in the exam. The assignments will not be graded, but will provide numerical answer checking for live feedback while working. Reference solutions, and an example grading rubric, will also be provided so that students can grade their own work. Live support during scheduled lecture hours via the forum and video links. Support for asking questions outside of the scheduled hours in the forum at any time. Final exam will have the same format, and assessment criteria, as the assignments. The course will make use of a mandatory mid-term assessment structure. In a nominal trajectory, the student will complete a closed and secured 1 hour midterm exam halfway through the course, and a 2 hour final exam at the end of the course. Should a retake be desired by the student, this will take place in a 3 hour exam.
Assessment	Nominal trajectory: 30% midterm exam (1h), 70% final exam (2h) Retake trajectory: 100% final exam (3h)

TN2545	Systems and Signals	6
Responsible Instructor	Dr. F.M. Vos	
Instructor	Dr.ir. D.J. Verschuur	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2A 2B	
Start Education	2A	
Exam Period	2A 2B 3A	
Course Language	English	
Expected prior knowledge	Complex numbers, elementary calculus, including derivatives & integrals, solving linear equations with few unknowns, geometric series, trigonometric & exponential functions. Create and analyze graphs of elementary functions	
Course Contents	This course covers the following topics: Systems and signals in continuous and discrete time; Basic deterministic signals; Convolution and properties of LTI systems; Fourier series and Fourier transforms with their properties; Filters in theory and practice: first- and second-order DE, Butterworth filters, Gaussian derivative filters; the estimation of spectra in a time window and the associated uncertainty relation; the DFT and FFT as practical tools; introduction to 2D Fourier transforms; modulation and demodulation techniques; the analytic signal and Hilbert transforms; sampling (Nyquist theorem), aliasing and reconstruction; laplace and Z-transforms with their ROC and properties; feedback systems and associated stability determination.	
Study Goals	The main learning goals of this course on Systems and Signals are to enable that a student can: 1. recognize and determine basic properties of systems including linearity, time invariance, impulse response, and determine their response based on convolution. 2. derive Fourier series coefficients and Fourier transform of elementary signals. 3. describe the effect of frequency-dependent filtering of discrete and continuous-time signals and relate this to linear difference/differential equations with constant coefficients. 4. analyze the effect of sampling of band-limited signals and reconstruct a continuous-time signal from the discrete samples. 5. perform amplitude-modulation to elementary signals and derive and analyze associated reconstruction methods. 6. assess properties of systems based on the Laplace and Z-transform.	
Education Method	Oral lectures by teachers, practical working classes with supervision	
Literature and Study Materials	A.V. Oppenheim and A.S.Willsky, with S.H.Nawab, Signals and Systems, Prentice-Hall, 2nd edition, Ch. 1 to 11 (obtainable in bookshop and VVTP). Reader with additional material en lecture notes/hand-outs.	
Assessment	There are 3 bonus tests during the teaching period before the exam takes place. They are optional and can give a student a bonus to the final grade (10% contribution per bonus test). If a bonus test grade (BG) is lower than the exam grade (EG) it will not contribute. Therefore, the final grade (FG) is computed as one of these options, depending on the bonus grades: $FG = 0.7 EG + 0.1 (BG1+BG2+BG3)$ $FG = 0.8 EG + 0.1 (BG1+BG2)$ $FG = 0.9 EG + 0.1 BG1$ $FG = 1.0 EG$ Bonus test results are only valid in the academic year in which they are obtained.	

TN2626	Statistical Physics	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr.ir. D. Lathouwers	
Instructor	Prof.dr. S. Stallinga	
Contact Hours / Week x/x/x/x	Octaal 7: 3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege Octaal 8: 4x 2 uur p/w hoorcollege 4x 2 uur p/w werkcollege	
Education Period	4A 4B	
Start Education	4A	
Exam Period	4A 4B 5AB	
Course Language	Dutch	
Expected prior knowledge	Thermodynamics, TN1201	
Course Contents	Microstates, Macrostates. Paramagnets, Einstein solids, ideal classical gasses. Entropy and temperature. Chemical potential. Free energy. Boltzmann distribution, partition function and Gibbs factor. Bose-Einstein and Fermi-Dirac statistics. Ideal quantum gases, applications of quantum statistics including Bose-Einstein condensation.	
Study Goals	Being able to explain the concepts of the topics mentioned under 'Course Contents' Being able to identify the relation between the different concepts in the theory of statistical physics Being able to solve elementary problems, using the standard procedures of the theory. Being able to solve advanced problems on the theory, using mathematical skills and physics insight. Being able to solve problems in statistical physics numerically; relate outcomes of computer programmes to the theory.	
Education Method	Lectures, two kinds of exercise classes (homework assignments and computer assignments). Exercise sessions with computer assignments are partly dedicated to Python. The Python sessions with computer assignments address practical exercises for numerical solutions to problems in statistical mechanics.	
Computer Use	Python will be employed in the studio classroom sessions.	
Literature and Study Materials	D. V. Schroeder, An Introduction to Thermal Physics, Oxford University Press (Oxford, UK, 2021). ISBN: 9780192895554 (Paperback) Note: also Hardcover and E-book editions are available (Oxford University Press) Alternatives: Older editions (out-of-print): 1. D. V. Schroeder, An Introduction to Thermal Physics, Addison Wesley Longman, International edition, Ch. 1-7. 2. D. V. Schroeder, An introduction to Thermal Physics, Daniel V Schroeder, Pearson New International Edition (this is a reprint of the first seven chapters of the Longman Edition) Lecture Notes on Statistical Physics, by J. M. Thijssen (to be found on Brightspace) Videos, which can be accessed directly from the Lecture Notes, by a PDF link or by scanning a QR code.	
Assessment	Written final exam (open-book, mandatory) Python exercises (mandatory) Midterm exam (open-book, optional) The exam (and midterm exam) will be open book, meaning that it is allowed to use the book An Introduction to Thermal Physics by Daniel V. Schroeder and Jos Thijssens lecture notes. Other documents or notes are not allowed. It is allowed to use a (graphical) calculator in the exam (and midterm exam). The Python exercises (P) count for 20 % to the final mark. The midterm exam counts for 35% provided that the result for the midterm (M) exceeds the mark for the final exam (F) and the mark for the final exam is at least 5. The weighting factor of the final exam therefore lies between 45 % (if the midterm exam mark was higher than the final exam mark of at least 5) and 80 % (the midterm exam was not done or the midterm exam mark was lower than final exam grade or the grade of the final exam is below 5). The final grade G therefore is given by: $G = 20\% P + 80\% F$, or $G = 20\% P + 45\% F + 35\% M$, provided that $M > F$ and $F \geq 5$. The same results apply for the retake exam. Results obtained for the Python exercises remain valid for one year; the result obtained in 2023-24 can therefore be re-used in 2024-25. Note that only the final grade of the Python exercises may be re-used.	
Remarks	The lectures will be given in Dutch (assignments in English).	

TN2786	Physical Transport Phenomena	6
Responsible Instructor	Prof.dr.ir. C.R. Kleijn	
Instructor	Dr.ir. M. Rohde	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	4A 4B	
Start Education	4A	
Exam Period	4A 4B 5AB	
Course Language	Dutch	
Required for	BSc Applied Physics	
Expected prior knowledge	Introduction to Thermodynamics TN1201, Differential equations WI2140TN	
Parts	part 1 : chapter 1 - 3.5.2 part 2 : chapter 3 - 5	
Course Contents	Conservation laws and Macroscopic and microscopic balance equations for mass, momentum and energy. Dimensionless numbers, dimensional analysis. Hydrodynamic forces on solid bodies. Convective and diffusive transport of mass, momentum and energy. Heat and mass transfer, Nusselt and Sherwood relations. Thermal radiation. Technical fluid dynamics. 1-dimensional laminar flows of Newtonian and non-Newtonian fluids.	
Study Goals	1. Acquiring basic knowledge of in the field of Transport Phenomena. 2. Acquiring insight in the role that transport phenomena play in nature, technology, sustainability and daily life, and the consequences for the possibilities and limitations of technical solutions for societal and sustainability problems. 3. being able to recognize and analyze the relevant mechanisms which lead to the transport of mass, momentum and energy, and being able to set up and solve the resulting balance equations. 4. being able to (quantitatively) solve balance equations in the field of transport phenomena, in relation to applications in physics, process technology, energy technology, sustainability, environmental technology and daily life, making use of (approximating) theoretical, analytical and numerical methods. 5. being able to judge the correctness of obtained and postulated solutions and theoretical equations (based on, amongst others, dimensional analysis, order of magnitude estimates, empiricism, and rational engineering arguments.)	
Education Method	2x4 weeks with 8 hours per week lectures and tutorials = 64 hours. 2x4 weeks with 8 hours per week individual study (theory and problems) = 64 hours. 2x20 hours preparation for midterm and final exam.	
Computer Use	Two numerical assignments require - Knowledge of Python programming - availability of a (laptop) computer	
Literature and Study Materials	Mandatory books: -Harry Van den Akker and Robert F. Mudde, Fysische Transport Phenomena - the Art of Balancing, 1st edition - Martin Rohde, Graphs, Tables and Formulas for Transport Phenomena (order a *hardcopy* via https://textbooks.open.tudelft.nl/textbooks/catalog/book/103) -H.E.A.Van den Akker, R.F. Mudde en Stammers, 200 Vraagstukken Fysische transportverschijnselen. All published by Delft University Press.	
Books	Recommended: R.B. Bird, W.E. Stewart and E.N. Lightfoot, Transport Phenomena, 2nd edition, John Wiley & Sons 2002.	
Assessment	Online or on-campus examination: - Mandatory exam part 1 (50%): chapter 1 - 3.5.2; minimum grade 4.75; - Mandatory exam part 2 (50%): chapter 3 - 5. Minimum grade 4.75; - passed when average grade $(=0.5 \cdot (\text{part1} + \text{part2})) \geq 5.75$. - No resit for part 1 and part 2. - Resit about the entire course (part 1 + part 2) (100%).	
Permitted Materials during Tests	For on-campus closed book exam: -Book: Martin Rohde, Graphs, Formulas and Tables for Transport Phenomena -Non-graphical, non-programmable calculator	
Tags	Challenging Energy Energy & Industry Fluid Mechanics Intensive Modelling Physics Process Sustainability Technology Thermodynamics Transport phenomena	
Studyload/Week	2x4 weeks with 8 hours per week lectures and tutorials = 64 hours. 2x4 weeks with 8 hours per week individual study (theory and problems) = 64 hours. 2x20 hours preparation for midterm and final exam.	

TN2953-P	Research Laboratory		6
Responsible Instructor	Dr. E. van der Kolk		
Education Period	1A 1B 2A 2B 3A 3B 4A 4B		
Start Education	1A 3A		
Exam Period	none		
Course Language	Dutch English		

TN3104WI	Probability and Statistics		3
Responsible Instructor	Dr. R.C. Kraaij		
Education Period	3A		
Start Education	3A 3B		
Exam Period	3B 4B		
Course Language	Dutch		

Year	2024/2025
Organization	Applied Sciences
Education	Bachelor Applied Physics

2e jaar Module 2024

TN2304	Quantum Mechanics 1	3
Responsible Instructor	S.C. Conesa Boj	
Instructor	Prof.dr. S. Groeblacher	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 4x 2 uur p/w werkcollege	
Education Period	3A	
Start Education	3A	
Exam Period	3A 4A	
Course Language	English	
Required for	Quantum 2, Quantum 3, Solid state physics	
Expected prior knowledge	Calculus, Differential Equations, Linear Algebra, Classical Mechanics	
Course Contents	Griffiths chapters 1-3. Including basic concepts in quantum mechanics, one dimensional Schroedinger equation, quantum states and formalism, etc.	
Course Contents Continuation	Intended Audience TN2304 is an introductory course in quantum mechanics, and is a required course for the second year program in Applied Physics. Textbook Introduction to Quantum Mechanics, 3rd Edition by David J. Griffiths and Darrell F. Schroeter, Cambridge University Press Edition, ISBN-13: 978-1107189638 Weekly Schedule Lectures There are three 2-hour lectures per week, each of which is followed by a one-hour werkcollege. Both the lectures and the werkcolleges will take place in person. For each lecture, there is assigned reading from the book, which should be read before the lecture. The assigned reading material is listed in the "Schedule" link on the sidebar. During the lectures, the instructors (Simon and Sonia) will present a summary of the material, work through examples, and answer questions. Werkcolleges Following each lecture, there will be a one-hour werkcollege, which will also take place in person to encourage active participation and direct interaction. During the werkcollege, you will solve homework problems related to the topic of the lecture, by yourself or in groups if you wish. Teaching Assistants will be present during the werkcollege to assist you with the homework problems. The homework for the tutorial sessions will be posted on Brightspace well in advance before the werkcollege. Students are expected to practice it before the werkcollege and therefore maximise the time for live questions. The corresponding solutions will be posted on Brightspace by the end of the day where each homework set is scheduled. Minitests Each week, there will be a one-hour minitest covering the material of that week, conducted in person. Both the minitests and the exam will be "closed book," meaning no supporting materials are allowed except for a pen, pencil, and a basic calculator. Each minitest will be composed by several questions and problems of the same level and style as those that will appear in the final exam. After the end of each minitest, we will post on Brightspace the corresponding solutions, this way student can assess how well they did. NO Formula sheets We will not provide any physics equations on the formula sheets with the exams and minitests (no Schroedinger equation, no commutation relations, etc.). You will be responsible for remembering the basic formulas and commutation relations. The formula sheets in the minitests and exam this year will contain only mathematical formulas as deemed necessary (specific integrals, double angle formulas, etc.). Final grade assessment On each test/exam, a number of points that will be indicated that are assigned to each subquestion. The resulting grade of each test or exam will be calculated as follows: zero points corresponds to a grade of 1.0, full points corresponds to a grade of 10.0, with a linear scaling between the two. Language The official language of the course is Dutch. Neither Sonia nor Simon are fluent in Dutch and only have a very basic level of Dutch proficiency. Both instructors will give the lectures in English. The book is written in English. The homework questions will be in English. The minitest and exam questions will be written in English, but all students are welcome and free to write their answers in Dutch should that be more comfortable. Some of the Teaching Assistants may be Dutch native speakers, but the others have very limited Dutch knowledge. Clustering with Quantum 2 Starting from 2015, Quantum I will now be "clustered" with Quantum 2 (TN2314), with implications for the requirements on your final grade in order to pass the course: contact your study advisor for more info.	
Study Goals	Learning Objective 1: Essential quantum phenomena	

The student will be use the uncertainty principle, quantum measurement, quantum tunnelling, and zero point energy to predict how objects behave differently in quantum mechanics than in classical mechanics.

Learning Objective 2: Observation and measurement in quantum mechanics

Given a specific wavefunction, the student will be able to calculate the probability of measurement outcomes, the expectation values of observables, and be able to write down the form of the wavefunction after a measurement.

Learning Objective 3: Eigenstates of time-independent Hamiltonians

The student will be able to find the eigenstates and eigenenergies for specific time-independent Hamiltonians, and use these eigenstates to calculate expectation values and the probability of measurement outcomes.

Learning Objective 4: Time dependence of the wave function and of expectation values

For a time-independent Hamiltonian and an initial wavefunction at $t=0$, the student will be able to calculate the time dependence of the wavefunction and of expectation values as a function of time.

Learning Objective 5: Scattering potentials

The student will be able write down the boundary conditions for the wavefunction for step-function 1-dimensional potentials. The student will be able to use these to calculate transmission and reflection coefficients.

Learning Objective 6: The uncertainty principle

The student will be able to use the generic uncertainty principle to calculate uncertainties in observables.

Learning Objective 7: Properties of wavefunctions and operators

The student will be able to explain why wavefunctions and operators in quantum mechanics need to satisfy specific mathematical requirements.

Learning Objective 8: Dirac notation

The student will be able to use Dirac notation to express quantum states and expectation values.

Learning Objective 9: Commutators

The student will be able to calculate commutators of quantum mechanical operators.

Learning Objective 10: Basis transformation

Given a wavefunction for a quantum state expressed in one basis (for example, position basis), the student will be able to transform this into a wavefunction in a different basis (for example, momentum basis).

Learning Objective 11: Matrix representation

The student will be able to calculate eigenfunctions, expectation values, and operators in vector and matrix representation.

Education Method

Books

Lectures, tutorials, homework

Introduction to Quantum Mechanics, 3rd Edition by David J. Griffiths and Darrell F. Schroeter, Cambridge University Press Edition, ISBN-13: 978-1107189638

<https://www-cambridge-org.tudelft.idm.oclc.org/nl/academic/subjects/physics/quantum-physics-quantum-information-and-quantum-computation/introduction-quantum-mechanics-3rd-edition?format=HB&isbn=9781107189638>

There are several "unofficial" modified versions of this book (the international edition from Pearson for example). These are also OK but often use different chapter section numbering, different homework question numbering, are missing some chapters, have often useless indices, and contains some errors. You will be personally responsible for correctly translating the section numbers and homework question numbers into the correct numbers from if you do not choose the correct version of the book.

Assessment

Each of the minitests will be allowed to count for up to 10% of your final grade if it increases your final grade. This means that you have nothing to lose by taking a minitest. This is to encourage you to take the minitests in order to get early feedback on how you are doing in the course.

As an example: suppose that your grades for the four minitests are (5.3 / 8 / 8.2 / 7.3) and your final exam grade is 7.1. Since three of your minitest grades are higher than your final exam grade, the weighting for the calculation of your final grade would be 0% Minitest 1, 10% for each of Minitest 2/3/4, 70% Final exam.

After calculating your final grade, it will be rounded off to the nearest half-point and then submitted to OSIRIS.

Final written exam: The exam will be "closed book" and will take place at the TU Delft campus.

Remarks

Kwantummechanica 1 en 2 vormen samen een module met een vijfenregeling, waarbij voor beide vakken geldt dat minimaal een 5 moet worden gehaald en het gewogen gemiddelde gelijk aan of groter dan 5,75 moet zijn.

TN2314	Quantum Mechanics 2	3
Responsible Instructor	T.H. Taminiau	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 4x 2 uur p/w werkcollege	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4B	
Course Language	Dutch	
Course Contents	kwantummechanica in drie dimensies, identieke deeltjes, toepassingen ontleend aan recent onderzoek.	
Study Goals	1) Deepened insight into the concepts of quantum mechanics and the differences with classical physics. 2) Gain insight into the wave functions of the hydrogen atom. What do they look like spatially and why? What is the relationship between shape of the orbitals and physical quantities such as energy and angular momentum. 3) Solving the Schrödinger equation using separation of variables. The techniques need to be applied to diverse situations and the cohesion between separation constants and quantum numbers needs to be understood. 4) The ability to apply operator techniques such as the commutation relations and ladder operators to solve quantum mechanical problems. 5) Understanding the relationship between the concept of identical particles and the symmetry of the wave functions and how this leads to the Pauli exclusion principle. 6) Understanding the periodic system of elements from a quantummechanica perspective. This includes understanding what quantum numbers derived for the H-atom are also applicable for many-electron atoms. 7) The student must understand how in quantum mechanics angular momenta add up and how this manifests itself in the energy levels of atoms/ions. 8) Understanding the quantum observable spin, and the ability to apply the quantum rules to calculate expectation values, measurement outcomes, and probabilities for measurement outcomes for measurements of spin.	
Education Method	hoor/werkcollege, wekelijkse toetsen	
Literature and Study Materials	D.J. Griffiths and D.F. Schroeter, Introduction to Quantum Mechanics, third ed. Cambridge University Press.	
Assessment	Written exam. Several intermediate minitests. Each mintest grade counts for 10% of the final grade, if and only if it improves the final grade.	

Year	2024/2025
Organization	Applied Sciences
Education	Bachelor Applied Physics

3e jaar BSc TN 2024

SC3011TN	Stochastic Signal Analysis	3
Responsible Instructor	Dr. P. Mohajerin Esfahani	
Contact Hours / Week x/x/x/x	0/0/0/0/10/0/0/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	Modeling of stochastic signals, correlation, spectra, estimation theory, estimating spectra and transfer functions, time series models, optimal filtering; examples focused on situations from technical physics.	
Study Goals	After the course, students can - describe and analyze stochastic signals in the time and frequency domain and they can use these insights to quantify measurement signals disturbed by noise; - use concepts such as auto- and cross-correlation as well as auto- and cross signal spectra; and estimate these quantities from finite series of measurements. - calculate the change of the correlation functions and spectra when stochastic signals are filtered by a linear dynamic system; - solve the inverse problem, viz. how they can calculate the forming filter from the stochastic spectra. - estimate the information-carrying components from disturbed measurement signals, - derive and use the linear optimal Wiener filter and Least Squares estimators to solve noise suppression problems.	
Education Method	Lecture with tutorial consisting of theoretical and python exercises	
Literature and Study Materials	Stochastic Processes for Physics Students: An Introduction, by C.S. Smith and M. Verhaegen. Course Reader 2021.	
Assessment	Written exam and homework. The final mark is a weighted average of the mark obtained on the exam (85%) and that on the compulsory homework (python) assignments (15%).	
Department	ME Department Delft Center for Systems and Control	

TN2412	Quantum Mechanics 3	3
Responsible Instructor	Prof.dr. A.F. Otte	
Instructor	Dr. R.H.M. Smit	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	3A	
Start Education	3A	
Exam Period	3A 4A	
Course Language	Dutch	
Expected prior knowledge		
Course Contents	Hydrogen atom (summary), angular momentum, spin. Time independent perturbation theory. Time dependent perturbation theory. Relativistic corrections, including kinetic and spin-orbit contributions. Approximation methods, density matrices.	
Study Goals	Being able to use introductory notions of quantum mechanics in order to discuss theoretically, and apply to computational problems, the following concepts: - the energy spectrum of the hydrogen atom and its wavefunctions - quantisation of the orbital angular momentum, spin angular momentum and their combinations - approximation methods for calculating energy spectra, including time-independent perturbation theory in its degenerate - relativistic corrections, including kinetic and spin-orbit contributions - selection rules - quantum dynamics within the time-dependent perturbation theory formulation - density matrix formalism	
Education Method	Theoretical and exercise classes	
Literature and Study Materials		
Assessment	Written exam	
Permitted Materials during Tests		

TN2421	Optics	3
Responsible Instructor	Dr. A.J.L. Adam	
Instructor	Dr. rer. nat. M. Menzel	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4B	
Course Language	Dutch English	
Expected prior knowledge	Analyse (WI1400TN). Inleiding Mechanica, golven en optica (TN1302). Elektromagnetism (TN2053). Introduction to waves (TN2345). Systems & Signals (TN2545).	
Course Contents	· Geometrical Optics (§2) - Fermats Principle - Lenses, Pupils, Mirrors - Ray Matrix · Optical Instruments (§3) - Eyes - Magnifying Glass - Camera - Microscope - Telescope · Polarisation (§4) - State of Polarisation - Polarisers, Dichroism, Birefringence - Retarders & Circular Polarisers - Matrix Representation - Scattering · Interference & Coherence (§5) - Two Waves Addition - Interference and Conditions - Spatial and Temporal Coherence - Fringes - Interferometers · Scalar Diffraction Optics (§6) - Huygens Principle - Fraunhofer Diffraction - Fresnel Diffraction - Angular and Spatial Resolution - Diffraction Grating - Fresnel Diffraction · Lasers (§7) - Lasers, Heat Source, Gas Source: unique properties - Amplification & Cavities - Problems of Laser Operation - Types of Lasers	
Study Goals	Upon successful completion of this course, students will be able to: 1. Solve optical systems in the domain of geometrical optics, both graphically and analytically. 2. Examine wave interference and diffraction patterns, considering principles of spatial and temporal coherence. 3. Compute the changes in the polarization state of light after passing through polarizing and birefringent materials using Jones calculus. 4. Explain optical phenomena and the working principles of optical systems.	
Education Method	Lectures and tutorials hours	
Literature and Study Materials	A book available on the Open TextBook at TUDelft is the reference for the course. You can find it here: https://textbooks.open.tudelft.nl/textbooks/catalog/book/42 The notes and slides from the lectures will be available on Brightspace. It contains all information needed for the course. Videos of experiments will be offered on Brightspace/youtube.	
Books	A book available on the Open TextBook at TUDelft is the reference for the course. You can find it: https://textbooks.open.tudelft.nl/textbooks/catalog/book/42 You can check Hecht "Principle of Optics" or Goodman "Introduction to Fourier Optics" by Joseph Goodman for further reading.	
Assessment	There is an individual written bonus exam (multiple choice) after the first half of the course, which is optional and gives a bonus of max. 1 point in 0.1 steps (BP). At the end of the course, there is an individual written final exam which counts for a maximum of 10 points (G). The final grade is the final exam grade added to the bonus point (G + BP), given between 1 and 10 in steps of 0.5. Students pass if they have at least 6 out of 10. Resit: individual written exam for 100% of the grade. The bonus point does not count for the resit exam. Attention will be made on the quality of the explanation/reasoning.	
Exam Hours	3 hours	
Permitted Materials during Tests	Calculator	

TN2844	Solid State Physics	6
Responsible Instructor	Dr. A.R. Akhmerov	
Instructor	Dr. T. van der Sar	
Contact Hours / Week x/x/x/x	2x 2 uur p/w hoorcollege 2x 2 uur p/w werkcollege	
Education Period	3B 4A	
Start Education	3B	
Exam Period	3B 4A 5AB	
Course Language	English	
Course Contents	* Vibrations in solids: density of states, Debye model, specific heat. * Electrons in metals: Drude model, Hall effect, free electron model, Fermi surface, density of states, specific heat. * Atoms and bonds: periodic table, orbitals, atomic bonds. * Electrons and phonons in 1D: reciprocal space, Bloch theorem, phonon band structure, tight-binding model, nearly-free electron model, effective mass, group velocity, density of states. * Lattices in higher dimensions: crystal structure, unit cells, basis, reciprocal space, Brillouin zone, X-ray diffraction. * Tight-binding and nearly-free electron models in 2D and 3D. * Semiconductors: charge balance, doping, semiconductor devices.	
Study Goals	In each specific topic, after following the course the students will be able to: * Vibrations in solids: - Describe the Debye model, explain its limitations - Calculate the density of states predicted by Debye model - Define heat capacity due to phonons, approximate it in low- and high temperature limits * Electrons in metals: - Write down the Drude model, use it to calculate conductance properties of metals - Explain the concept of Fermi sea and Fermi surface - Calculate the density of states of electrons with power law dispersion - Define and calculate the electron heat capacity * Atoms and bonds: - Describe the properties of atomic orbitals - Translate a description of molecular orbital structure into a tight-binding Hamiltonian, obtain its spectrum * Electrons and phonons in 1D: - Write down the equations of motion of a 1D chain of atoms and springs, apply the Bloch theorem, calculate normal modes using the eigenvalue problem - Formulate the tight-binding Hamiltonian for an electron system in 1D and derive the corresponding dispersion and density of states - Determine the effective mass for a given dispersion * Lattices in higher dimensions: - Classify a given crystal structure in terms of a unit cell and basis. - Construct the reciprocal lattice and first Brillouin zone corresponding to a real-space lattice - Formulate the structure factor for a crystal and predict its diffraction properties for a certain scattering vector * Tight-binding and nearly-free electron models in higher dimensions - Explain the relation between filling factor and the conductive properties of materials - Apply the nearly-free electron model for a 1D crystal and derive the dispersion - Write down and solve the tight-binding and nearly-free electron models given a general description of the system in 2D and 3D * Semiconductors: - Use charge balance equations to calculate charge carrier densities (holes & electrons) and chemical potential depending on doping levels and temperature - Apply the approximations necessary to derive the properties of intrinsic and extrinsic semiconductors - Describe the functioning of basic semiconductor devices	
Education Method	Lectures and exercise seminars	
Literature and Study Materials	Steven H. Simon, The Oxford Solid State Basics (Oxford University Press, 2013); lecture notes (https://solidstate.quantumtinkerer.tudelft.nl).	
Assessment	3 bonus minitests and a final exam, with the grade determined according to the formula $F + (M_1 + M_2) * (10 - F) / 100$, with F being the exam grade, and M_1 and M_2 the best two minitest grades. If the final exam score is 0, the minitests give 2 bonus points, and if the final exam score is 10, the minitests give 0 bonus points. The minitests count towards the resit score if the minitest score is higher than that of the resit exam.	
Tags	Modelling Physics	

TN2983	Bachelor Thesis	12
Responsible Instructor	Dr. R.H.M. Smit	
Contact Hours / Week x/x/x/x	N.v.t.	
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	Exam by appointment	
Course Language	Dutch	
Study Goals	1. Theoretical knowledge and understanding (theoretical knowledge and application of theory) 2. Method and scientific approach (responsibility, communication, literature study, critical attitude and time planning) 3. Competence in doing research work (extension/generation of methods, new ideas, performing experiments and scientific significance) 4. Reporting (quality of report, independent in writing and usefulness of report) 5. Presentation & defence (quality of presentation, handling questions and depth of argumentation in oral defense) 6. Competences (level of English, independency, interpersonal skills, creativity and open-mindedness)	

TN3100	Individual research project (var EC)	0
Responsible Instructor	Dr. R.H.M. Smit	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	Dutch English	

WM0318TN	Argumentation and Philosophy of Science	3
Module Manager	Dr. M. Astola	
Instructor	Dr. M. Astola	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	see Dutch version	
Study Goals	see Dutch version	
Education Method	see Dutch version	
Assessment	see Dutch version	

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